

Henry Demarest

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EDUCATION

Stanford University

Palo Alto, CA

Master's Degree in Aerospace Engineering

Sept. 2025 – June, 2027

- **GPA:** 3.96
- **Awards:** 2025 National Science Foundation Graduate Research Fellowship Program (NSF GRFP)

Yale University

New Haven, CT

BS in Mechanical Engineering (ABET), BS in Computer Science

Aug. 2021 – May 2025

- **GPA:** 3.98
- **Awards:** Summa Cum Laude, Phi Beta Kappa, Tau Beta Pi, L.C. Lichty & E.O. Watters Prize in Mechanical Engineering, Matthew Isakowitz Fellowship, Robert E. Lewis Award, Wesley M. Oler Prize
- **Non-Technical Activities:** First-Year Counselor, Vice Chair of Yale ASME, MechE Peer Mentor, Engineering Tour Guide, Yale Club Running, DPops Orchestra, Backpacking Trip Leader, Intramural Secretary

Irvington High School

Irvington, NY

High School Degree

Sept. 2017 – June, 2021

- **Weighted GPA:** 4.58, **SAT:** 1600
- **Awards:** Valedictorian (2021), National Merit Semifinalist (2021), U.S. Presidential Scholar Semi-Finalist (2021)

WORK EXPERIENCE

Relativity Space

Long Beach, CA

GNC Engineering Intern

June 2026 - Sept 2026

- Fully owned the development, tuning, and analysis of a novel flight algorithm to be used on TerranR's ascent
- Maintained and improved a monte carlo simulation case related to pad drift on launch

GNC Simulation Engineering Intern

May 2025 - Aug 2025

- Developed an improved architecture for modeling rocket engine dynamics during flight
- Integrated an aerodynamics model for plume-body interactions during stage separation into simulation

GNC Performance Engineering Intern

May 2024 - Aug 2024

- Led key trade studies focused on optimizing vehicle trajectory via different loads-informed throttle controllers
- Implemented automated feature to improve link between monte carlo data and trajectory optimization framework

Astrobotic Technologies

Pittsburgh, PA

Lunar Surface Systems - Mechanical Engineering Intern

May 2023 - Aug 2023

- Designed test stands to subject components (solar arrays and dust covers) to cryogenic/vacuum environments
- Iterated upon the design of a spacerless-MLI thermal enclosure to enable spacecraft survival of the lunar night

PROJECTS

Yale Undergraduate Aerospace Association

New Haven, CT

Executive Board - President

May 2023 - May 2024

- Led the executive board and engaged with sponsors, administrators, aerospace companies, and other stakeholders
- Hosted inter-college conferences, industry recruitment events, volunteering opportunities, and technical workshops

Project Liquid: Chief Engineer (2023-24), Co-President (2022-23), Propulsion Lead (2021-22) *Sept 2021 - June 2025*

- Led the development of a regeneratively-cooled liquid rocket using nitrous oxide and ethane as propellants
- Managed the design, manufacturing, and testing of a custom spark ignitor
- Designed a coaxial shear injector using a stacked manifold design as Propulsion Lead

CubeSat Team – Attitude Determination and Control System (ADCS) Lead (2022-23)

Sept 2021 - June 2023

- Led the ADCS Subteam to prepare a CubeSat for launch as part of NASA's CubeSat Launch Initiative
- Designed and implemented an algorithm to control CubeSat's active stabilization efforts using magnetorquers

Heat Shield Tile Impact Testing

New Haven, CT

Mechanical Engineering Senior Design Student - Nanoprobe Group

June 2024 - May 2025

- Worked with NASA Ames Research Center to conduct impact testing on spacecraft heat shield tiles
- Designed, built, and tested a custom impact hammer, spring-powered silder, and penumatic pellet launcher

Quadcopter Ergodic Controller Implementation

New Haven, CT

Computer Science Senior Design Student - Intelligent Autonomy Lab

June 2024 - May 2025

- Wrote code to implement a decentralized ergodic search algorithm in a CrazyFlie drone's embedded controller
- Conducted preliminary testing in the lab's drone cage to validate drone's operation

TEACHING

Testive/A+ Tutoring

Remote

Tutor

September 2025 - Present

- Work with mostly high school students seeking assistance with coursework or SAT/ACT test preparation

Independent Tutoring - Wyzant

Remote

Tutor

July 2025 - Present

- Work one-on-one with learners ranging from elementary school students to full-time engineers
- Tutored subjects including math, physics, computer science, test preparation, United States history, and more

Yale University - ENAS 772 (Introduction to Embedded Robotic Systems)

New Haven, CT

Undergraduate Learning Assistant

Jan 2025 - May 2025

- Held office hours with students to go over course material for Introduction to Embedded Robotic Systems
- Graded homework assignments and provided feedback for students

Yale University - CS223 (Data Structures & Programming Techniques)

New Haven, CT

Undergraduate Learning Assistant

Sept 2022 - Dec 2023

- Supported students in CPSC 223 (Data Structures & Programming Techniques) and helped organize the class
- Held weekly office hours, teaching about the C programming language, data structures, and debugging strategies

RESEARCH

Stanford NavLab

Palo Alto, CA

Master's Student Researcher

Sept 2025 - Present

- Developed SLAM (simultaneous localization and mapping) benchmarking capabilities for the lab
- Working on team to improve autonomous lunar navigation systems

Yale Nanoprobe Group

New Haven, CT

Undergraduate Student Researcher

Jan 2023 - May 2025

- Conducted research on the microstructural effects of aging MP35N, a nickel-chromium superalloy
- Used nanoindentation, scanning electron microscopy, x-ray diffraction, and other advanced analysis techniques

Yale Intelligent Autonomy Lab

New Haven, CT

Undergraduate Student Researcher

Sept 2024 - May 2025

- Developed a decentralized, multi-agent ergodic search implementation to run on Crazyflie nano-quadcopters
- Worked with C firmware to efficiently calculate ergodic trajectories onboard drones

Cube Satellite Detumbling Research Project

New Haven, CT

Undergraduate Student Researcher

May 2022 - Aug 2022

- Designed and simulated a detumbling algorithm for a 2U CubeSat with limited sensors and actuators
- Used TRIAD algorithm for attitude determination and B-dot controllers paired with magnetorquers for control

Machine Learning Research Project with Mentors at IBM

Yorktown, NY

High School Student Researcher

January 2019 - May 2021

- Assessed the performance of reinforcement learning algorithms with a bipedal walking task using Python code
- Earned 1st Place at NYS Science and Engineering Fair (NYSSEF) and attended Regeneron ISEF competition

VOLUNTEERING

Stanford Young Astronauts Program

Woodside, CA

Student Volunteer

Sept 2025 - Present

- Teach monthly lessons at local elementary schools on topics around aerospace engineering

Yale Pathways

New Haven, CT

Event Organizer for Aerospace Association Event

Sept 2023 - June 2024

- Organized volunteering Yale Undergraduate Aerospace Association events in collaboration with Yale Pathways
- Taught New Haven students about CAD, rocketry, and electronics with hands-on lessons

Code Haven

New Haven, CA

Student Mentor

Sept 2021 - May 2022

- Taught coding skills to New Haven middle school students, explaining complex ideas in fun, understandable ways

Fuller Center for Housing

Yonkers, NY

Chapter Founder, Volunteer Supervisor, Intern

June 2018 - Aug 2021

- Repaired and remodeled local homes to be used as affordable housing in local communities
- Volunteered for up to 35 hrs/week at this non-profit during the summers and founded a chapter at my high school

TECHNICAL SKILLS

Programming Languages: C, C++, D, Java, Python, MATLAB, Julia, Rust

CAD: Solidworks, Autodesk Inventor, Fusion360, OnShape

Manufacturing: Lathe, Mill, CNC, 3D-Printer, Hand Tools

Software: Project Management (Confluence/Jira), Collaboration (GitLab, GitHub, Solidworks PDM), Microsoft Office